

Technical Document #865: Retrieval Augmented Generation - Comprehensive Overview

Technical Document #865: Retrieval Augmented Generation - Comprehensive Overview

Query decomposition breaks complex queries into simpler sub-queries. It improves retrieval quality for multi-faceted questions.

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. It grounds LLM responses in factual, retrieved documents.

Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Semantic search finds documents based on meaning rather than exact keywords. It uses embedding similarity to match queries with relevant content.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Key Takeaways:

- Context window optimization selects the most relevant information to fit within the LLM's context limit.
- Chunking splits documents into smaller segments for embedding.
- Retrieval-Augmented Generation (RAG) combines information retrieval with text generation.